Persistence

Lecture 9 - CMSE 890

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Tues, Sep 30, 2025

Goals

Goals for today:

- Persistent homology

Section 1

Last time

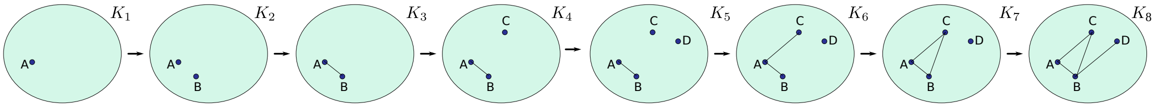
Last time: Simplicial Complex Filtration

Definition

A filtration $\mathcal{F} = \mathcal{F}(K)$ of a simplicial complex K is a nested sequence of its subcomplexes

$$\mathcal{F} : \emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$$

$\mathcal{F}$ is called *simplex-wise* if $K_i \setminus K_{i-1}$ is empty or a single simplex.

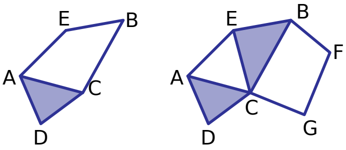


Betti curve

The p-th Betti curve is a function

$$\begin{aligned}\beta_p(\mathcal{F}) : \mathbb{Z} &\to \mathbb{Z}\\ i &\mapsto \beta_p(K_i) = \mathrm{rk}(H_p(K_i))\end{aligned}$$

Induced map



Given a simplicial map $f : K \to L$, the induced map on homology is defined by

$$\begin{array}{rccc} f_* : & H_p(K) & \to & H_p(L) \\ & [\alpha] & \mapsto & [f_\#(\alpha)] \end{array}$$

Section 2

Persistence

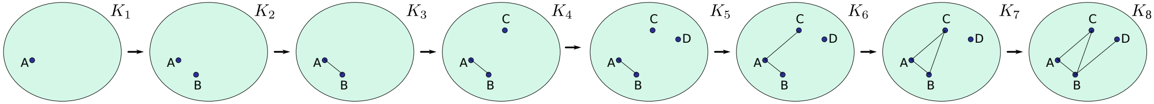
Persistence module

Given a filtration $\mathcal{F}$, the p-dimensional persistence module is

$$H_p(\mathcal{F}) : 0 = H_p(K_0) \to H_p(K_1) \to H_p(K_2) \cdots \to H_p(K_n) = H_p(K)$$

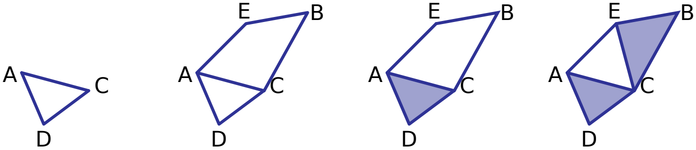
with maps $h_p^{i,j} : H_p(K_i) \to H_p(K_j)$ induced by inclusion.

What's going on algebraically?



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

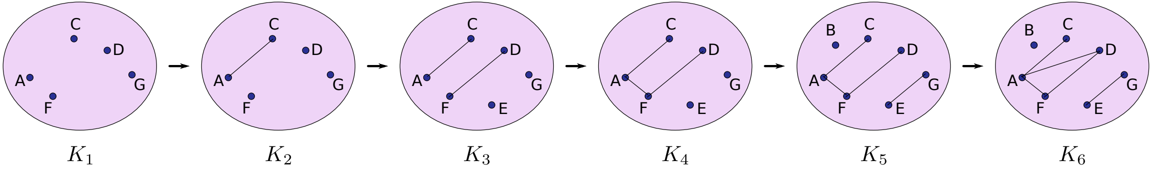
Another example



K_0 K_1 K_2 K_3

$$H_1(K_0) \xrightarrow{f_*} H_1(K_1) \xrightarrow{g_*} H_1(K_2) \xrightarrow{h_*} H_1(K_2)$$

Tryit: Figure out where the generators go



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6)$$

The pth persistent homology groups

$$H_p(K_0) \to H_p(K_1) \to H_p(K_2) \to \cdots \to H_p(K_i) \to \cdots \to H_p(K_j) \to \cdots \to H_p(K_n)$$

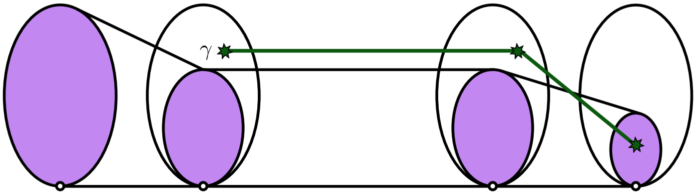
The pth persistent homology groups are the images induced by inclusion:

$$H_p^{i,j} = \mathrm{Im}(H_p(K_i) \to H_p(K_j)).$$

The pth persistent Betti numbers are the ranks

$$\beta_p^{i,j} = \mathrm{rank} H_p^{i,j}$$

Birth and Death



Birth

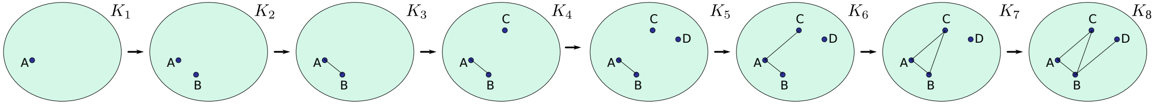
A class $\gamma \in H_p(K_i)$ is born at K_i if it is not in $H_p^{i-1,i}$

Death

That class γ dies entering K_j if merges with an older class.[a] Specifically, if $h_p^{i,j-1}(\gamma) \notin H_p^{i-1,j-1}$ but $h_p^{i,j}(\gamma) \in H_p^{i-1,j}$.

[a] Warning: The book's definition is different

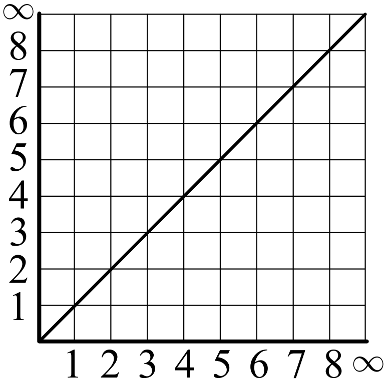
When are generators born and die?



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

Elder rule and persistence diagram

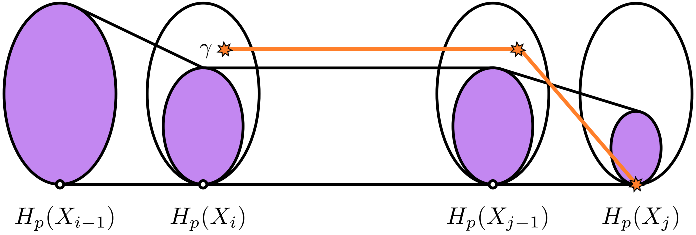
When two classes merge, the younger is the one that "dies."



Section 3

Dey Wang version of persistence classes

Birth and Death: Book version



Birth

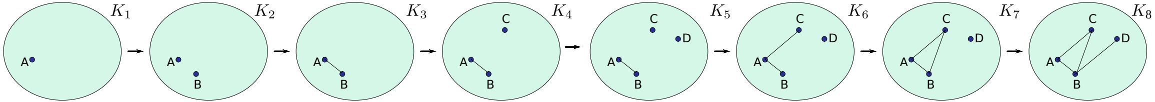
A class $\gamma \in H_p(K_i)$ is born at K_i if it is not in $H_p^{i-1,i}$

Death

A class γ dies entering K_j if $\gamma \in H_p(X_{j-1})$ is not trivial, but $h_p^{j-1,j}(\gamma) = 0$.[a]

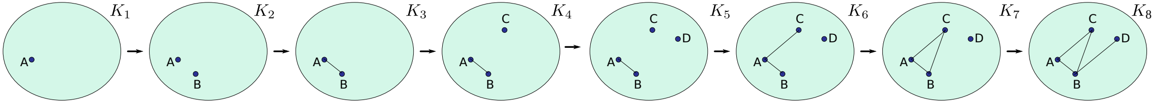
[a] Warning: In this version, not all classes die!

Lets check where (lots of but maybe not all) elements go



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

Chosing a different basis



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

Book definition: which birth time pairs to the death time?

Let $[c]$ be a p-th homology class that dies entering X_j. Then, it is born at X_i if and only if there exists a sequence

$$i_1 \leq i_2 \leq \cdots \leq i_k = i$$

for some $k \geq 1$ so that

- $[c_{i_\ell}]$ is born at X_{i_ℓ} for every $\ell \in \{1, \cdots, k\}$.
- $[c] = h_p^{i_1, j-1}([c_{i_1}]) + \cdots + h_p^{i_k, j-1}([c_{i_k}])$
- $i_k = i$ is the smallest possible value among any sequences have the above two properties.

Counting classes

$$0 \to H_p(K_1) \to H_p(K_2) \to H_p(K_3) \to \cdots \to H_p(K_n) \to 0$$

- Attach 0 vector space at the end
- Associate $n+1$ to $a_{n+1} = \infty$
- $\beta_p^{i,j}$ counts classes born **before** i and dying **after** j
- How to get the number of classes born **at** i and dying **at** j???

Persistence pairing function

For $0 < i < j \leq n+1$, define

$$\mu_p^{i,j} = \begin{cases} (\beta_p^{i,j-1} - \beta_p^{i,j}) - (\beta_p^{i-1,j-1} - \beta_p^{i-1,j}) & i < j < n+1 \\ \beta_p^{i,j} - \beta_p^{i-1,j} & i < j = n+1 \end{cases}$$

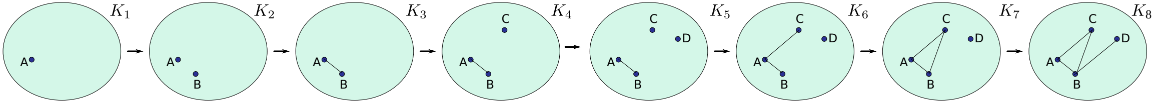
Persistence of a class

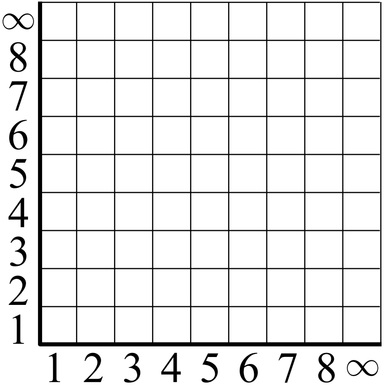
For $\mu_p^{i,j} \neq 0$, the persistence $Pers([c])$ of a class $[c]$ that is born at X_i and dies at X_j is defined as $Pers([c]) = a_j - a_i$. When $j = n+1$ with $a_{n+1} = \infty$, $Pers([c]) = \infty$.

Persistence diagram

The persistence diagram $Dgm_p(\mathcal{F})$ (also written $Dgm_p(f)$) of a filtration induced by a function f is obtained by drawing a point (a_i, a_j) with non-zero multiplicity $\mu_p^{i,j}$ $(i < j)$, on the extended plane where the points on the diagonal $\Delta = \{(x, x) \in \mathbb{R}^2\}$ are added with infinite multiplicity.

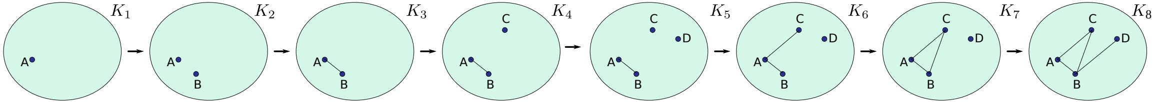
TRYIT: Example

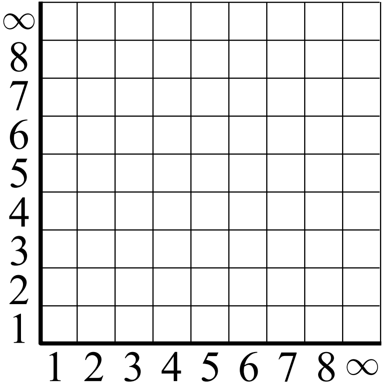




Determine $\beta_0^{i,j}$ for all pairs $i \leq j$ for this example

TRYIT: Example

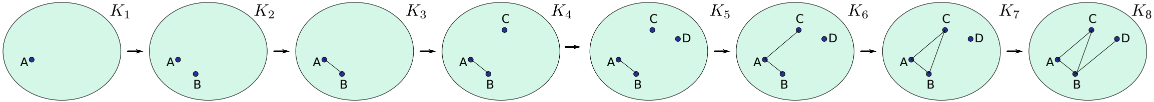


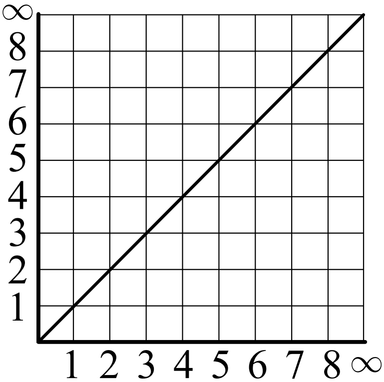


Determine $\mu_0^{i,j}$ for all pairs $i < j$ for this example

$$\mu_p^{i,j} = \begin{cases} (\beta_p^{i,j-1} - \beta_p^{i,j}) - (\beta_p^{i-1,j-1} - \beta_p^{i-1,j}) & i < j < n+1 \\ \beta_p^{i,j} - \beta_p^{i-1,j} & i < j = n+1 \end{cases}$$

TRYIT: Persistence diagram for this example





Plot the 0-dimensional persistence diagram

Homework

What is the 0-dimensional persistence diagram of the filtration below?